



Datasheet

WLD-521-R1 Water Leak Detection Module

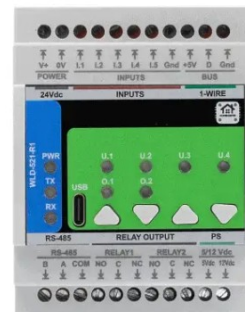
Part No.: WLD-521-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board + Relay Extension)

Controller: RP2350A



Functional Overview

The WLD-521-R1 is a modular water leak detection and water metering + I/O module for monitoring and automation. It provides 5 opto-isolated digital inputs, 2 SPDT relay outputs, 1-Wire sensor support, and optional 4 buttons / 4 user LEDs for local control and indication. The module connects to a PLC, MicroPLC, MiniPLC, SCADA, or other Modbus master as a Modbus RTU slave over RS-485 and is configured via USB-C Web Serial (WebConfig).

Board	Function
MCU Board	RP2350A controller, RS-485 communication, USB-C configuration interface, power conversion
Field Board	5 isolated digital inputs, isolated sensor power rails, 1-Wire interface
Relay Extension	2 × SPDT relay contact outputs

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (W25Q32JV, 32 Mbit)
Power Input	24 V DC nominal
Input Protection	1 A fuse, reverse polarity diode, TVS surge suppression, EMI filtering
Main Logic Supply	Buck regulator 24 V → 5 V, 3.3 V LDO regulator
File system	LittleFS persistent configuration storage
Document revision	DS-WLD-521-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
Digital Inputs	5 × isolated 5 V DC discrete inputs
Max Input Voltage	5 V DC continuous
Relay outputs	2 × SPDT (NO / C / NC); 3 A @ 250 VAC per contact (module limit)
Isolated +12 V Output	12 V isolated rail, 2 W; max 167 mA theoretical (≤ 150 mA recommended)
Isolated +5 V Output	5 V isolated rail, 1 W; max 200 mA theoretical (≤ 150 mA recommended)
1-Wire	1 × protected 1-Wire interface (+5 V, DATA, GND)
User Interface	4 buttons; 9 LEDs (Power, 4 user, RX, TX, Relay1, Relay2)
RS-485	half-duplex Modbus RTU, not galvanically isolated
USB	USB-C, 5 V logic, ESD protected
Typical Power Consumption	3 W typical / 5 W maximum (both relays energised, isolated sensor supplies fully loaded)
Modbus defaults	Address 3, 19200 baud, 8N1
Operating temperature	0 °C to +40 °C
Storage temperature	−10 °C to +55 °C
Relative humidity	0–90 % RH, non-condensing
Ingress protection	IP20 (inside cabinet only)
Installation	Indoor control cabinet only; not for outdoor or exposed installation
Maximum altitude	2000 m
Pollution degree	2
Dimensions	70 × 90.6 × 67.3 mm (L × W × H)
DIN width	4 modules (≈ 70 mm)
Mounting	35 mm DIN rail
Enclosure	PC/ABS industrial enclosure
Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire cross-section	0.2–2.5 mm ² (AWG 24–12)
Tightening torque	0.4–0.6 Nm
Net weight	TBD

Gross weight	TBD
Pack size	140 × 125 × 94 mm (L × W × H)

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Typical module power consumption: 3 W typical / 5 W maximum (both relays energised, isolated sensor supplies fully loaded).

Top Terminal Block (1-11)

Front view – terminals numbered from left to right.

Position	Marking	Signal	Description
1	V+	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	I.1	INPUTS	Input 1
4	I.2	INPUTS	Input 2
5	I.3	INPUTS	Input 3
6	I.4	INPUTS	Input 4
7	I.5	INPUTS	Input 5
8	Gnd	INPUTS	Isolated input return (GND_ISO)
9	+5V	BUS	1-Wire supply (logic domain, not isolated)
10	D	BUS	1-Wire data
11	Gnd	BUS	1-Wire ground (logic)

Bottom Terminal Block (1-11)

Front view – terminals numbered from left to right.

Position	Marking	Signal	Description
1	B	RS485	RS-485 data -
2	A	RS485	RS-485 data +
3	COM	RS485	RS-485 signal reference
4	NO	RELAY1	Relay 1 normally open
5	C	RELAY1	Relay 1 common
6	NC	RELAY1	Relay 1 normally closed
7	NO	RELAY2	Relay 2 normally open

8	C	RELAY2	Relay 2 common
9	NC	RELAY2	Relay 2 normally closed
10	5Vdc	PS	Isolated sensor rail +5 V
11	12Vdc	PS	Isolated sensor rail +12 V

Installation, Environmental & Mechanical

Specification	Details
Terminal Type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire Cross Section	0.2–2.5 mm ² (AWG 24–12)
Tightening Torque	0.4–0.6 Nm

Specification	Details
Operating Temperature	0 °C to +40 °C
Storage Temperature	–10 °C to +55 °C
Relative Humidity	0–90 % RH, non-condensing
Ingress Protection	IP20 (inside cabinet only)
Maximum Altitude	2000 m
Pollution Degree	2

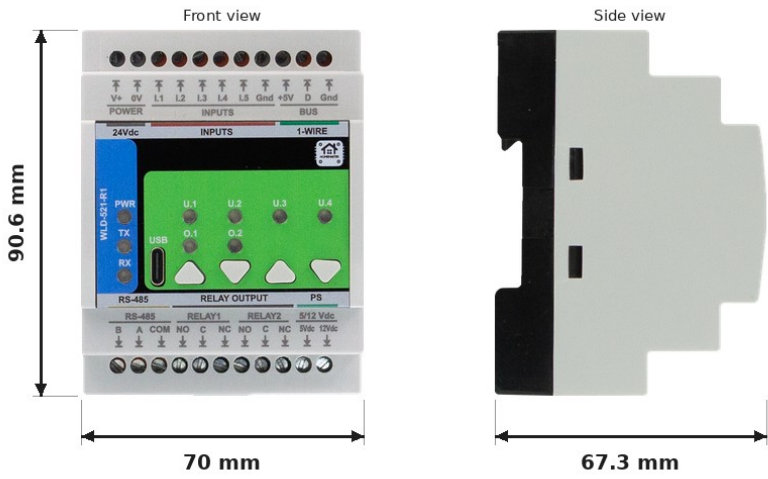
Specification	Details
Dimensions	70 × 90.6 × 67.3 mm (L × W × H)
DIN units	4 modules (≈ 70 mm)
Net weight	TBD
Gross weight	TBD
Pack size	140× 125 × 94 mm (L × W × H)
Mechanical drawing	Mechanical drawing: front and side view, dimensions in mm
Notes	All dimensions shown in millimeters unless stated otherwise
Mounting	35 mm DIN rail (EN 50022)
Enclosure Material	PC/ABS, UL94-V0

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.

WLD-521-R1

All dimensions in mm



Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

RS-485 Cable

- Twisted pair recommended (120 Ω characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.
- Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

Compliance & Certifications

The WLD-521-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

Safety Notice

The power input and signal terminals are SELV circuits. Relay output terminals may switch hazardous voltage up to 250 VAC.

Adequate insulation and separation must be ensured during installation. Installation by qualified personnel only.

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